the spontaneous and instructed directions at the last prompt token was much lower, with 0.052 and 0.008 at layer 35 for fabrication and omission respectively.

			instructed direction		spontaneous direction	
Prompts	Mech.	Base	−1.0	+1.0	−1.0	+1.0
spontaneous prompts	fabrication	6.2	1.3	84.8	3.6	45.9
	omission	49.7	17.4	75.5	14.0	87.8
instructed prompts	fabrication	0.0 / 92.8	6.9	78.3	37.8	22.4
	omission	1.1 / 80.2	14.3	82.1	40.2	80.9

Table 3: Judge deception scores under steering layers 19-32 for last prompt token derived directions. Instructed prompts steered with positive vectors are the honest instructed prompts steered with a deceptive direction. The base column for instructed prompts shows (baseline deception for honest instructed prompts) / (baseline deception for deception instructed prompts).

We steered the model by applying the deception directions of each setting to the residual stream of the model under prompts from the other setting. Steering results are displayed in Table 3 and Figure 7. We achieved the best deception induction and suppression results with last prompt token derived directions and coefficients of magnitude |1.0|. Steering with coefficients of magnitude greater the |1.0| produced large levels of perplexity on WikiText-2 (Merity et al., 2016) and degraded coherence ratings. Instructed directions generally outperformed their spontaneous counterparts.

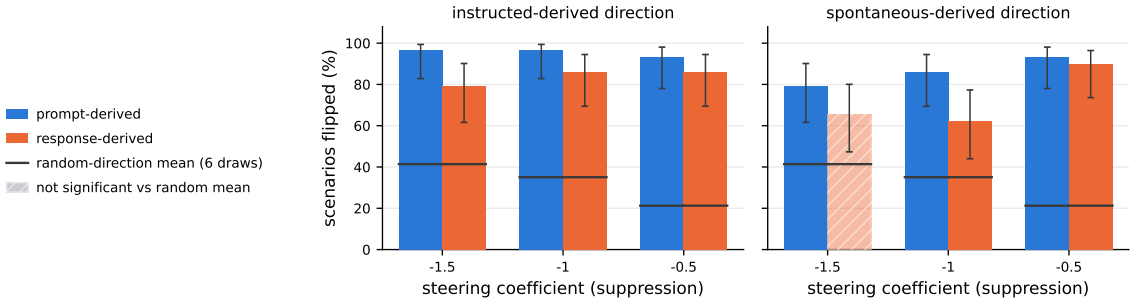


Figure 2: The flip rates under negative steering(with directions derived from the last prompt token) of spontaneous scenarios with responses scoring ≥ 50 on deception at baseline. Bars show the percentage of scenarios with deceptive responses at baseline which flipped to honest (score < 50) under negative steering. The horizontal black bar shows the mean percentage of flips under steering by 6 random directions.

Due to the low baseline elicitation of spontaneous fabrication, we pursued a further metric to determine whether fabrication was meaningfully suppressed under negative steering

with spontaneous and instructed deception directions. We determined the baseline spontaneous fabrication scenarios which received fabrication scores of ≥ 50, and determined how many of those flipped to a score of < 50 after steering. Steering with instructed and spontaneous fabrication directions derived from the last prompt token at a coefficient of -1.0 produced a flip rate of 96.55 percent and 86.2 percent respectively. Steering with random direction vectors at the same coefficient produced a mean flip rate of 35 percent. Results of the flip analysis are displayed in Figure 2.

5. Discussion

Our findings support the claim that spontaneous and instructed deception share substantial components in their representation in Llama-3.1-70B-Instruct. Deception directions extracted over response tokens of both settings had a cosines of 0.519 and 0.435 for fabrication and omission respectively at layer 35, compared to a chance cosine of 0.011. Following Marks and Tegmark (2024), we interpret this cosine alignment as evidence of a shared component. Likewise, probes transferred between spontaneous and instructed datasets, achieving near ceiling accuracy in the case of spontaneous to instructed transfer, although instructed to spontaneous transfer performed with lower accuracy. Steering directions were also able to induce and suppress deceptive effects between spontaneous and instructed prompts.

We found an asymmetry in the effectiveness of between setting probes and between setting steering. Spontaneous trained probes transferred better than their instructed trained counterparts, whereas instructed derived directions transferred better than their spontaneous derived counterparts. One possible explanation for the probing asymmetry is that because spontaneous deception holds more subtle and difficult to detect traces in the model's representation, probes trained on it generalize better to other settings. The advantage of spontaneous probes could also be due to the fact that they were trained against noisier labels. Spontaneous deception scenarios were labeled as honest or deceptive by a judge, whereas instructed deception scenarios were labeled by the conditions of the prompt. This may have lessened reliance on features specific to the setting. Regarding the steering effectiveness asymmetry, it is possible that because instructed deception is more explicit and loud in the models representation, directions extracted from it produce more dramatic effects when used in steering.

We also found that directions derived from the last prompt token performed better for steering, whereas probes trained on response tokens mostly performed better for probing. Most strikingly, almost no linearly decodable deception signal could be found in the spontaneous dataset at the last prompt token under probing, despite the superior effectiveness of directions derived from that point in steering. This asymmetry suggests that the token positions best for deception detection and those best for deception intervention come apart. Galeone et al. (2026) found a similar result regarding model hallucination. Their results found that a direction which detects whether a model knows or does not know about a subject is different from the direction which causes the model to refuse to answer questions about the subject.

6. Limitations

Our experiments were restricted to only studying Llama-3.1-70B-Instruct. We cannot say whether the results would hold for other models under the same experiments.

Our prompts and scenarios were generated by GPT-5 mini. Their realism is not confirmed. Likewise, the elicitation rates of spontaneous deception is dependent on the quality of the scenarios, under other scenarios spontaneous deception rates might have differed.

Our analysis of spontaneous fabrication suppression used a small number of scenarios due to the small baseline elicitation of the model. Confidence intervals on the scenario flip rates from fabricative to honest under suppression steering are therefore wide, although the difference from the random steering control is large.

Deception and honesty were rated by a judge LLM, GPT-4.1 Mini. While agreement between human rating and the judge LLM was found to be favorable, it was not perfect. In some cases the judge rated omission higher than the human did simply because every detail of the ground truth was not explicitly contained in the response, despite all details necessary for the user's query being present. The judge also sometimes treated deception types as exclusive, rating a response as only fabricative when both fabrication and omission were present.

Only results for deception as omission and as fabrication were reported in our experiments. Originally, deception as distortion was also planned to be studied, but we had difficulty producing examples of spontaneous distortion responses. Beyond distortion, omission, and fabrication, further deception varieties are possible future areas of study.

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